A

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination : Semester Final — Semester: **Fall 2025**
Duration: **2 Hours** — Full Marks: **70**

CSE421 / EEE465 : Computer Networks
Answer **Sections A, B and C** as per instructions given. (**Pages: 4**)
Figures in the right margin indicate marks.

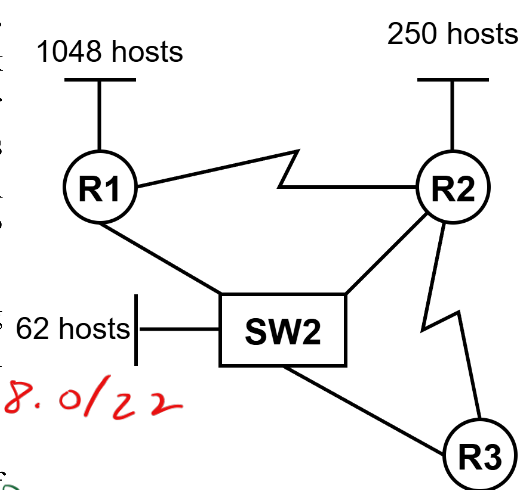
Name:	**ID:**	**Section:**

SECTION A [All questions of this section are **MANDATORY**] - **40 MARKS**

Q1 [CO3] A network consists of two subnets, referred to as Network A and Network B. Network A and Network B are consecutive subnets within the same major address block (i.e., there are no other subnets between them). Network A supports 1022 +2 hosts. A PC in Network A is assigned the first usable IP address of its subnet. **4 + 2 + 10**

This PC sends an IP packet with the following addresses: Source IP address: 172.16.8.1 Destination IP address: 172.16.13.255. A = 172.16.8.0/22

The packet is received by all devices in Network B.



I. **Determine** the network (subnet) address of Network B. 172.16.12.0/23 (2)

II. **Determine** the maximum number of hosts that Network B can support. (2) 510

III. **Given** the IP block 10.192.128.0/18. Apply VLSM to identify the sub-network addresses for each subnet present in the figure shown. The number of hosts given in the topology only includes the end devices. Next Page

Q2 A network-layer PDU of 4680 bytes, including a 20-byte header, is fragmented for transmission over a link with an MTU of X bytes. The difference between the fragment offset values of two consecutive fragments is 75. The fragment offset of the first fragment is 0. **3 + 3 + 4 + 3**

[CO3] I. **Calculate** the value of MTU (X). $(75 \times 8) + 20 = 620$

[CO3] II. **Calculate** the fragment offset of the 4th fragment. $3 \times 75 = 225$

[CO3] III. **Calculate** the data size of the last fragment.

[CO2] IV. If the DF value of the original datagram was set, how was the packet sent to its destination? Sends as whole / MTU Discover (3)

$$\frac{4680 - 20}{600} = 7.7$$

$$\therefore \text{last} = (4680 - 20) - 7 \times 600 = 460$$

1049 A
251 B
65 C
2 D
2 E

10.192.128.0/18

10.192.128.0/21 2
(A)

10.192.136.0/21

10.192.136.0/24 2
(B)

10.192.137.0/24

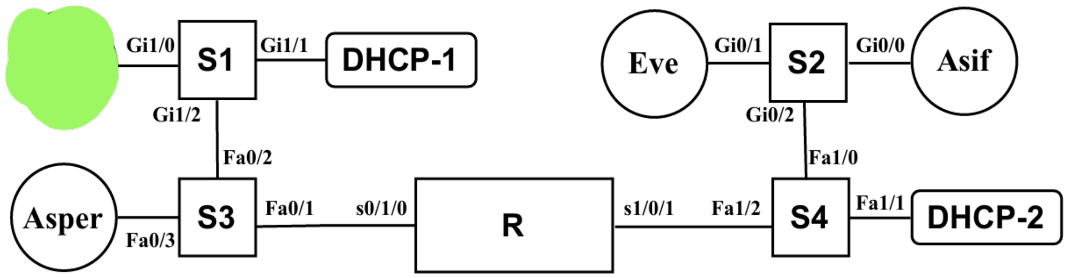
10.192.137.0/25 2
(C)

10.192.137.128/25

10.192.137.128/30
(D) 2

10.192.137.132/30
(E) 2

Q3 2 + 3 + 3 + 3
[CO2]



Saba and Eve have just connected to their networks. All switches have empty MAC tables; MAC-entry aging time is 2 minutes. Consider the MAC address of each host to be its name (e.g., ASIF).

(2)
0.0.0.0 **I.** Saba's PC sends its first DHCP message to obtain an IP address. What source IP address is used in this packet?

(1) Source MAC (2) **II.** Asper connects at the same time as Saba and sends a DHCP request. How can DHCP-1 distinguish Saba's request from Asper's request even though both requests are broadcast?

(1) fail = DORA via Relay (1) **III.** Eve has used an address from DHCP-2 for 5 hours. Her renewal attempt to DHCP-2 fails, but she then successfully obtains a new address from DHCP-1. Briefly explain how Eve is able to obtain an IP address from DHCP-1.

IV. Asper intends to send a packet to Eve. Since Eve is on a different subnet, Asper must send the packet to Router R. Name the specific link-layer requests and responses that are exchanged between Asper and Router R to make this possible?

(2) ARP Request + reply (1)

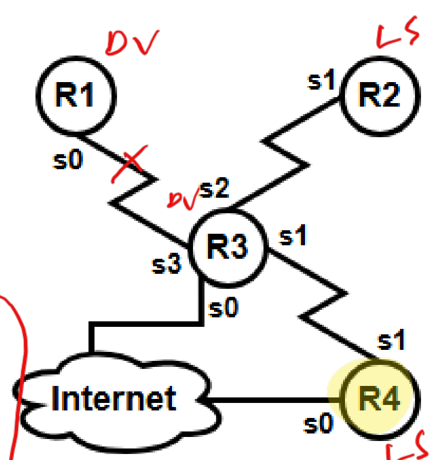
END OF SECTION A

[CO3] **SECTION B** [Answer **ANY TWO out of THREE** in this section] **- 12 MARKS**

Q4 Given a topology below. Each router has a LAN connected to its f0 interface. 3 + 3

I. **Configure** a non-recursive static route to R2 LAN from R4.

II. Assume R4 is routing to all unknown routes to the internet via s0. **Configure** a floating static route for this route.



Device	Int.	IP	Network
R1	f0	.1	10.10.10.0/24
	s0	.1	10.14.10.0/30
R2	f0	.1	10.10.11.0/24
	s1	.1	10.15.10.0/30
R3	f0	.1	10.10.12.0/24
	s0	.2	10.17.10.0/30
	s1	.2	10.16.10.0/30
	s2	.2	10.15.10.0/30
	s3	.2	10.14.10.0/30
R4	f0	.1	10.10.13.0/24
	s0	.1	10.18.10.0/30
	s1	.1	10.16.10.0/30

(1) (2)
ip route 10.10.11.0 255.255.255.0 s1
— 0.0.0.0 0.0.0.0 s1/10.16.10.2
(1) (2)

Q5 **I.** **Expand** the following IPv6 Addresses in their Hexadecimal form. 3 + 2

S2: 0/0 → Asif (1) | S4: 1/0 → Asif (1)
0/1 → Eve (1)

A. FE80::1A2B FE80:0000:0000:0000:0000:0000:0000:1A2B

B. 2001:DB8:1234::10 2001:0DB8:1234:0000:0000:0000:0000:0010

C. ::25 0000:0000:0000:0000:0000:0000:0000:0025

II. A PC sends a packet to 224.10.10.5, but only some PCs receive it. **Explain** why. Multicast

3

Q6 Using the given diagram in Q3, answer the following questions. **3 + 3**

I. Asif wants to send an IP packet to Eve, but does not yet know Eve's MAC address. What frame does Asif generate (source and destination MAC), and how do the switches S1, S2, S3, and S4 process this frame?

Src: Asif
Dest: FF:FF...
S2, S4 = Brd.
S1, S3 = nothing

II. After Eve replies to Asif, **show** the MAC address tables of all the switches.

END OF SECTION B

[CO2] **SECTION C [Answer ANY THREE out of FIVE in this section] - 18 MARKS**

Q7 **3 + 3**

I. Apart from limiting the number of hops, explain how the Time to Live (TTL) field helps prevent network problems. What would happen to the Internet if the TTL field did not exist? Trace.

II. An attacker discovers a network with 200 active devices. He sends one ICMP Echo Request in such a way that the target victim receives 200 ICMP Echo Replies. How is it possible? DDoS → Smurf / Magnification

Q8 Suppose Router R1, R3 in Q4's topology are using the Distance Vector Routing Algorithm, while routers R2 and R4 are using the Link State routing Protocol. Suddenly, the link between R1 and R3 fails. **3 + 3**

No. others = LS.

I. **Explain** whether this link failure will cause the count-to-infinity problem.

II. Once the link between R1 and R3 fails, all routers have only their directly connected networks in their routing tables. **Explain** the reason.

- R3 = DV = Breaking LS.
- R1 & R3 not communicating

Q9 Assume Host A & D inside a LAN trying to communicate with a remote host B through an edge router that performs PAT. The edge router has 220.20.20.17/30 in its PAT pool. **3 + 3**

A packet from host A to host B leaves with source IP 192.168.2.105, source port 52157, destination IP 6.6.6.6, and destination port 344. → 220.20.20.17:52157

A packet from host D to host B leaves with source IP 192.168.2.106, source port 52157, destination IP 6.6.6.6, and destination port 344. → 220.20.20.17:52158

I. **Identify** the source IP and port address of the packets sent by host A and host D when they leave the edge router for host B.

II. Suppose a remote host C on the Internet wants to initiate new communication with host A. **Explain** why or why not this would be possible with the current setup. If not, how can it be made possible? Port FWD

Q10 Scenario 1: An organization needs IPv6 devices to communicate with IPv4 devices during the transition period. **1 + 2 + 3**

Scenario 2: A service is deployed using the same IPv6 address on multiple routers. Clients always reach the nearest router automatically.

	I. **Name** the IPv6 transition technique used in Scenario 1. NAT ① **II.** What IPv6 addressing mechanism is used in Scenario 2? Is there any difference between this mechanism and multicast addressing? Anycast. Anycast = nearest ③ **III.** IPv6 does not have an Options field like IPv4. Since the IPv6 base header is fixed at 40 bytes, **mention** how IPv6 carries additional information when required. Extencion headers ③	
Q11	Rahim is learning about link layer communication. He notices that sometimes a frame sent using a MAC address reaches only one device, while at other times it reaches a group of devices. 8th bit -> 0=U, 1=M ③ **I.** Rahim wants to understand how to determine whether a given MAC address is unicast or multicast. **Explain** how he can identify this from the MAC address. **II.** His friend remarked that MAC addresses are hierarchical and portable. **Justify** whether his remark about MAC addresses is correct. Incorrect. Flat & Portable ① ②	**3** + **3**

END OF SECTION C

==================== **THE END** ====================

Why don't packets ever get lost at home?
They always know their default gateway.

B

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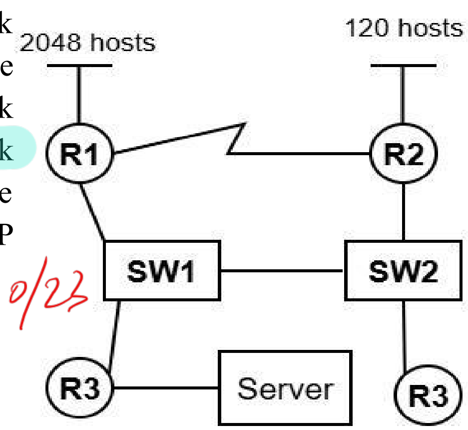
Name:	**ID:**	**Section:**

SECTION A [All questions of this section are **MANDATORY**] - 40 MARKS

Q1 [CO3] A network consists of two subnets, referred to as Network X and Network Y. Network X and Network Y are consecutive subnets within the same major address block (i.e., there are no other subnets between them). Network X supports 510 hosts. A PC in Network X is assigned the first usable IP address of its subnet. This PC sends an IP packet with the following addresses: — 4 + 2 + 10



Source IP address: 192.168.20.1 X= 192.168.20.0/23
Destination IP address: 192.168.23.255
The packet is received by all devices in Network Y.

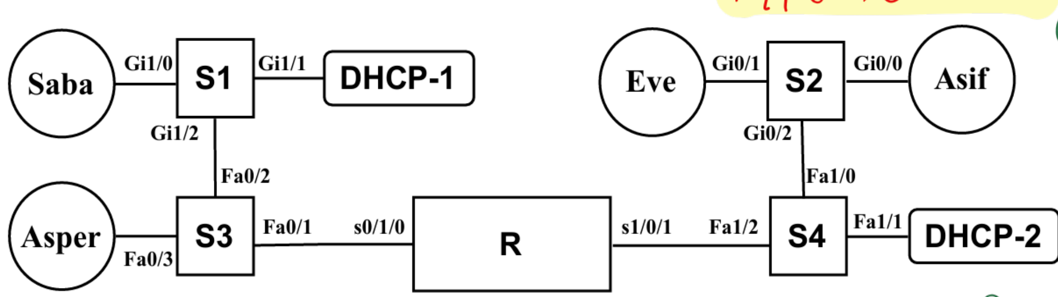
Y=192.168.22.0/23

I. **Determine** the network (subnet) address of Network Y.
510 II. **Determine** the maximum number of hosts that Network Y can support.
III. Given the IP block 190.132.0.0/19. **Apply** VLSM to identify the sub-network addresses for each subnet present in the figure shown. The number of hosts given in the topology only includes the end devices. next page.

Q2 A network-layer PDU containing 4000 bytes of data, and a 20-byte header, is fragmented for transmission over a link with an MTU of X bytes. The difference between the offset values of two consecutive fragments is 85. The fragment offset of the first fragment is 0. — 3 + 3 + 4 + 3

[CO3] I. **Calculate** the value of MTU (X). 85×8+20=700
[CO3] II. **Calculate** the fragment offset of the 5th fragment. 4×85=340
[CO3] III. **Calculate** the size of the last fragment.
[CO2] IV. **Mention** why the fragments are reassembled only at the destination, not in the routers.
MTU related ...

Q3 [CO2] — 2 + 3 + 3 + 3



$\frac{4000}{680} = 5.88 \rightarrow$ Last $= 4000 - 5 \times 680 + 20$
$= 620$

2049 121 4 2 1

A B C D E

500 9
1 10
2 11
4 12

190.132.0.0/19

190.132.0.0/20 (A) 2

190.132.16.0/20

190.132.16.0/25 (B) 2

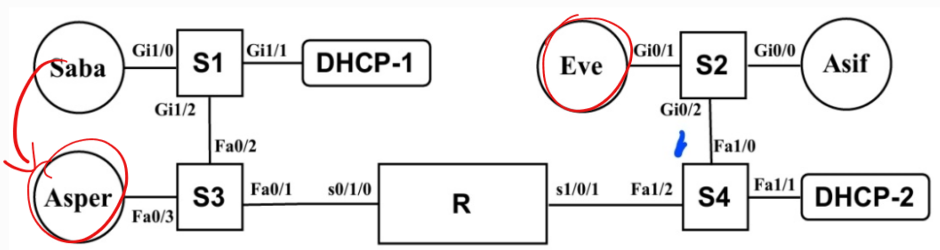
190.132.16.128/25

190.132.16.128/29 (C) 2

190.132.16.136/29

190.132.16.136/30 (D) 2

190.132.16.140/30 (E) 2



Asper and Asif have just connected to their networks. All switches have empty MAC tables; MAC-entry aging time is 3 minutes. Consider the MAC address of each host to be its name (e.g., SABA).

I. Asif's PC sends its first DHCP message to obtain an IP address. What destination IP address is used in this packet? 255.255.255.255 (2)

II. Eve connects at the same time as Asif and sends a DHCP request. How can DHCP-2 distinguish Asif's request from Eve's request? (1) Source MAC (2)

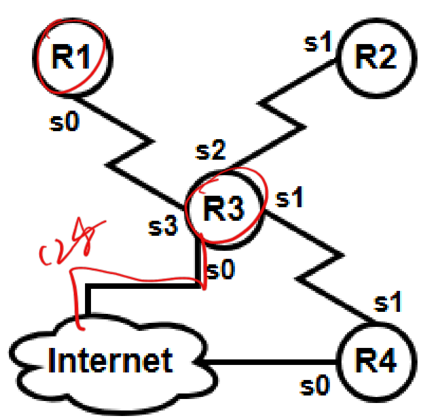
III. When Asper connects, DHCP-1 is powered off. Will Asper be able to join the network automatically? If not, state one possible configuration change that would allow Asper to obtain an IP address. (1) No. Setup relay. (2)

IV. Eve intends to send a packet to Asper. Since Asper is on a different subnet, Eve must send the packet to Router R. **Name** the specific link-layer requests and responses that are exchanged between Eve and Router R to make this possible. ARP Request (3)

END OF SECTION A

[CO3] **SECTION B** [Answer **ANY TWO out of THREE** in this section] **- 12 MARKS**

Q4 Given a topology below. Each router has a LAN connected to its f0 interface. 3 + 3



Device	Int.	IP	Network
R1	f0	.1	10.10.10.0/24
	s0	.1	10.14.10.0/30
R2	f0	.1	10.10.11.0/24
	s1	.1	10.15.10.0/30
R3	f0	.1	10.10.12.0/24
	s0	.2	10.17.10.0/30
	s1	.2	10.16.10.0/30
	s2	.2	10.15.10.0/30
	s3	.2	10.14.10.0/30
R4	f0	.1	10.10.13.0/24
	s0	.1	10.18.10.0/30
	s1	.1	10.16.10.0/30

I. **Configure** an efficient route to R1 LAN from R3.

(1) ip route 10.10.10.0 255.255.255.0 s3 (2)

II. Assume R3 is routing to all unknown routes to the internet via s0 with AD of 124. **Configure** a recursive backup static route for this route. 0.0.0.0 0.0.0.0 10.16.10.1 ?124 (1)

Q5 **I.** **Shorten** the following IPv6 Addresses. 3 + 3

A. 2001:0DB8:0000:0000:0000:FF00:0000:8329 (1) 2001:DB8::FF00:0:8329 (1)

B. FE80:0000:0010:0100:0202:B3FF:FE1E:8329 (1) FE80::10:100:202 ....

C. 2607:F8B0:0000:0800:0000:0000:0000:200E 2607:F8B0:0:800::200E

II. A PC sends a packet to 127.0.0.1, but the packet doesn't even leave the PC. However, the purpose of sending the packet has been fulfilled. **Explain** why. (3) Loopback / Virtual

Q6 Using the given diagram in Q3, answer the following questions. 3 + 3

I. Saba wants to send an IP packet to Asper and already knows Asper's MAC address. What frame does Saba generate (source and destination MAC), and how do the switches S1, S2, S3 and S4 process this frame? S: Saba, D: Asper. (1)

S1: 1/0 -> Saba
1/2 -> Asper (1)
S3: 0/2 -> Saba
(1) 0/3 -> Asper.
S2, S4 = Nothing (1)
S1, S3 = Broadcast (1)

II. After Asper replies to Saba, **show** the MAC address tables of all the switches.

END OF SECTION B

[CO2] **SECTION C [Answer ANY THREE out of FIVE in this section] - 18 MARKS**

Q7 I. A datagram network allows packets to take different paths and does not require call setup before sending data. A virtual circuit network uses a fixed path and maintains state information in routers. **Explain** briefly why a datagram network can be more robust than a virtual circuit network when some network links or routers fail. — 3 + 3

3

II. A corporate web server suddenly loses connectivity. The network team finds that the server's incoming bandwidth is fully used by thousands of ICMP Echo Requests coming at the same time from many ordinary home computers located in different countries. **Identify** the type of attack used.

3

DDos → Zombie

Q8 Suppose Router R1, R2, R3 and R4 in Q4's topology are using Distance Vector Routing Algorithm. Suddenly, the link between R1 and R3 fails. Then routers R2, R3 and R4 keep advertising increasing hop counts to unreachable networks. This is a major problem "X" in DVR. — 3 + 3

1 Count-to-infinity | Solution: Split Horizon / do not advertise / poison reverse 2

I. **Explain** problem "X" and its solution.

II. **Justify** whether this problem would occur in the Link State Routing Algorithm.

1 No, topology map 2

Q9 Assume Host A & D inside a LAN trying to communicate with a remote host B through an edge router that performs PAT. The edge router has 210.10.10.9/30 in its PAT pool. — 3 + 3

A packet from host A to host B leaves with source IP 192.168.2.105, source port 60255, destination IP 6.6.6.6, and destination port 344. 210.10.10.9:60255 1

A packet from host D to host B leaves with source IP 192.168.2.106, source port 60255, destination IP 6.6.6.6, and destination port 344. 210.10.10.9:60256 2

I. **Identify** the source IP address and port number of the packets sent by hosts A and D when they leave the edge router for host B.

II. Host A runs a Minecraft server on port 25565 with port forwarding to 192.168.2.105:25565. If Host C (on the same LAN) also wants to host on port 25565, can the same forwarding rule be created? **Explain** why or why not.

1 No. forwarding is based on port 2

Q10 A network administrator observes the following IPv6 addresses configured on different devices in a campus network — 2 + 2 + 2

Device A - FE80::1A2B Link Local 1

Device B - 2001:DB8:1234::10 Global 1

Device C - FC00:ABCD::25 Unique local

Device D - FF02::1

I. **Identify** the type of IPv6 address used by Device A, B, and C.

II. Which of these addresses cannot be routed by routers, and why? A & C → local / private 1 1

III. **Identify** the field that was introduced in the IPv6 header but was not present in the IPv4 header. Next header / flow label 2

00000010

Q11	Sunbeam finds a device with MAC address 02:1A:2B:3C:4D:5E. He thinks it is a factory-assigned MAC address, but his little sister says the address was manually set. **I.** Is his sister correct? **Justify** your answer by analyzing the first byte of the MAC address. 1 = local = sister correct 3 **II.** Later, Sunbeam wonders if MAC addresses are unique and portable, why are they not used for global Internet routing like IP addresses? Give two reasons why MAC addresses are not used for global Internet routing. Flat, Binded with device like a serial no. 3	**3 + 3**

END OF SECTION C

===================== **THE END** =====================

Why don't switches gossip?

They forward frames silently.